

# The Influence of Psychological Capital on Employees' Innovative Behavior: Mediating Role of Employees' Innovative Intention and Employees' Job Satisfaction

SAGE Open  
July-September 2021: 1–14  
© The Author(s) 2021  
DOI: 10.1177/21582440211040809  
journals.sagepub.com/home/sgo  


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## Abstract

The study investigates the impact of psychological capital on the employees' innovative behavior through the mediating effect of employees' job satisfaction and employees' innovative intention in the small and medium enterprises (SMEs) sector of Saudi Arabia. A sample of 204 respondents participated from various enterprises working without restricting specific sectors to check employees' common behavior in multiple sectors. The data and hypotheses testing analysis were made with the partial least squares–based structural equation modeling (PLS-SEM). The study revealed that psychological capital positively affects employees' job satisfaction, innovative behavior, and innovative intention. Furthermore, the employees' job satisfaction also positively correlated with the employees' innovative behavior, while there was no connection between the employees' innovative intention and the employees' innovative behavior. Concerning the indirect relationships, the findings revealed that employees' job satisfaction played a partial mediating role between psychological capital and the employees' innovative behavior. However, the employees' innovative intention did not mediate the relationship between the psychological capital and the employees' innovative behavior. These findings suggest the importance of psychological capital in influencing the innovative behavior of employees. Hence, there is a need to continue developing it among employees to ensure a better output.

## Keywords

psychological capital, employees' job satisfaction, employees' innovative behavior, Saudi Arabia, SMEs

## Introduction

Due to globalization and continuous rapid business changes, many business enterprises have emerged to offer various products and services in the market, intensifying stiff competition in the market and forcing many to leave due to their inability to compete and sustain. Enterprises attempt to produce unique products and services immutable by others and value added to ensure a share in the market. Good performance is the objective of every company, which requires solving problems, introducing new products and services, and taking advantage of newly available opportunities to improve organizational effectiveness (Rego et al., 2012). Therefore, there is a need to engage employees in thinking, designing, and implementing products and services. Engaging employees in such a strategy would require companies to work on individuals' specific personal characteristics and enhance their creativity and innovation.

Employees with innovative behavior and creativity skills play a crucial role in an enterprise's growth, development, and sustainability processes. These internal motives

have been classified under the umbrella of positive psychology theory developed by Seligman and Csikszentmihalyi (2000). Later, other aspects related to positive psychology were investigated. As a result, the positive psychological capital concept was introduced by Luthans, Youssef, and Avolio (2007), categorizing the psychological capital into four constructs: hope, self-efficacy, optimism, and resilience. It is also known as the individual positive psychological state that helps individuals familiarize themselves with their environment and stress, develop their competitive advantages, and advance their well-being (Tang et al., 2019). Furthermore, it is also considered an imperative intangible asset and reflects the employees' positive mental energy.

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Skilled and knowledgeable employees are considered valuable assets for companies and organizations (Fritz et al., 2011). As a result, concerned enterprises must consider necessary measures for strengthening their employees' innovative behavior. Furthermore, as psychological capital is regarded as a positive state of mind for employees, it can be improved by providing suitable training and specific development, allowing them to have valuable capital (Luthans et al., 2005).

Employees' encouragement and support are reflected positively on their performance (Seligman & Csikszentmihalyi, 2000). In general, members and leaders should have the ability to adapt to the rapid changes in the work environment to ensure sustainable development in their organizations (Sreitzer et al., 2012). Part of this change and development should include the so-called employees' psychological capital. It is believed that individuals with the capability to develop psychological capital features proactively are expected to have a higher confidence level, achieve success in challenging tasks, and obtain optimistic views on overcoming the problems faced currently or in the future (Luthans, Youssef, & Avolio, 2007), supported by authentic leadership to positively influence their activity (Rego et al., 2012). Employees' creativity and organizational innovation are destined to stagnate without the supervisor's positive support as there is a relationship between leadership, employees' psychological capital, and behavior (Wu & Chen, 2015).

Individuals' behavior may be affected by either positive or negative aspects of external contexts; however, they are more responsive to the external one. There is a requirement to concentrate more on developing positive psychological capital as it is considered a strategic resource that impacts organizations' internal work performance (Newman et al., 2014). The development of novel and valuable ideas about products, practices, services, or procedures has become increasingly crucial for the survival and competitiveness of organization these days (Shalley et al., 2009), and this will not happen unless there is a deep focus on the employees' creativity and innovation. This is possible with the assistance of the employees' psychological capital as it plays a vital role in enhancing employees' innovative behavior, intention, and attitude. It also strengthens the individuals' level of confidence and makes them more satisfied with their workplace.

In Saudi Arabia, the government has made many attempts to focus on the development and growth of the non-oil sectors, especially the infrastructures and the young human capital, to ensure the economic growth, development, provision of job opportunities, and benefit from the rich demographic dividends there (Ranjan et al., 2013). For that, in its long-term plans, particularly the so-called Saudi vision 2030, the government focused on improving the contribution of the small and medium enterprises (SMEs) sector and making its contribution reach 35% instead of 20% of the gross domestic product (GDP). The government has developed many initiatives for supporting this sector, including providing financial

and technical support and incubation services for those enterprises. However, it is believed that focusing only on economic and technical support might not produce the desired result. Achieving maximum output and an innovative performance would require creating an encouraging, supportive, and satisfying environment for SMEs sector employees to ensure creativity and innovation. Along with providing a creative and delightful environment, it is also essential to work on the employees' inner abilities and strengthen them by developing their psychological capital and directing it toward achieving better performance.

Given that, and due to the importance of the development of human capital worldwide and particularly in Saudi Arabia, it is noted that a limited number of researches were carried to investigate its effect on various constructs. For example, Alkahtani et al. (2020) revealed in their study that there is a positive connection between psychological capital, workplace well-being, and employees' engagement. Another study of Abosaif (2018) reported that psychological capital predicts the faculty members' quality of life at Altaif University. The relationship between psychological capital and the perceived organizational support on career adaptability was also examined, and the results reported a significant connection (Ibrahim & Amari, 2018). Finally, Idris and Manganaro (2017) disclosed no significant association between psychological capital with job satisfaction and organizational commitment and also reported that psychological capital influences Saudi women teachers in Saudi Arabia. Extant literature indicates the importance of psychological capital in motivating the employees to be more engaged in the job and have a better quality of life and workplace well-being. We, therefore, build on this principle as it is to be observed that only a few studies have been conducted on the role of psychological capital and the employees' innovative behavior in general and particularly with relation to SMEs. Hence, this fills the available research gap by examining the influence of psychological capital on the SMEs employees' innovative behavior and considering the job satisfaction and innovative intention as mediators.

The study is organized into eight sections. Following the introduction, the theoretical framework is discussed, followed by the research methodology of the study. The fourth section presents the data analysis and testing of hypotheses. Finally, the discussion, implications, limitations, and future research are then demonstrated.

## **Theoretical Background and Hypotheses Development**

### *Psychological Capital*

The increased competition and other economic changes globally and particularly in Saudi Arabia due to the continuous decline in the oil price have led Saudi Arabia to implement several reforms on various parts of its economy to

ensure more diversification, sustainability, innovation, productivity, and better performance. Accordingly, a long-term plan, namely, the vision 2030, was developed to meet those recent challenges and make the necessary adjustments. Vision 2030 focused on providing essential support to various sectors in the country, including the SMEs that play a crucial role in creating new job opportunities and reducing the unemployment rate. The 2030 plan aims to increase the SMEs' contribution in the GDP from 20% to 35% by 2030, the sector that hosts about 950,000 enterprises categorized between micro to small and medium and provides more than one million jobs. In addition, the government of Saudi Arabia has implemented several initiatives such as offering the necessary training and support for improving the performance of the employees and also vital funds for encouraging individuals to start their innovative income-generating activities, which may ultimately benefit the overall performance of the economy (Khan & Alsharif, 2019).

Accordingly, we believe that improving the performance of SMEs in Saudi Arabia can rely on the financial and technical support and other regulative issues provided. However, we also need to look at the effect of different factors that may impact the individuals' behavior and lead to positive and better performance. For example, ensuring a happy life for employees contributes to better performance and productivity (Seligman & Csikszentmihalyi, 2000). It also helps develop novel and valuable ideas about products, practices, services, or procedures that ensure the survival and competitiveness of an enterprise (Shalley et al., 2009). However, securing an acceptable level of happiness for employees requires providing an adequate environment in addition to increasing their confidence, improve their optimism, resilience, and hope (Luthans, Youssef, & Avolio, 2007). Therefore, we believe that employees in the SMEs sector can be encouraged to work more effectively and innovatively through developing their internal traits and strengths. Thus, the study is based on the positive psychology theory developed by Seligman, who selected it as a theme for his presidency period of the American Psychological Association (Seligman & Csikszentmihalyi, 2000). The approach is contrary to the practices in the past that concentrated on mental illness, thinking negatively, and encouraging making a happy life with a high level of positivity. Overall, the theory focuses on three main issues: positive emotions, positive individual traits, and positive institutions. These three issues focus on making an individual happy with his life and strengthening their relationship with the community. The theory also suggests a few factors that need to be encouraged as they lead to a better and happy life for individuals. These factors include, for example, having a solid network with other individuals in the society such as friends, family, spouse, colleagues, networks, organizations, and spirituality (Day, 2010).

The theory was later on investigated in various streams, including the organizational behavior field. Luthans, Avolio, et al. (2007) developed the psychological capital model as

they felt the need to establish a higher level construct model to represent the individuals' psychological capital. Accordingly, the concept of positive psychological capital was developed (Luthans, Youssef, & Avolio, 2007). Psychological capital is made up of four constructs, namely, self-efficacy, hope, optimism, and resilience. It is believed that it positively impacts individuals' behavior and provides them a solid background and a pathway for it. According to Luthans, Youssef, and Avolio (2007), psychological capital is defined as an "individual's positive psychological state of development." They further explained that psychological capital

is characterized by (1) having confidence (self-efficacy) to take and put in the necessary effort to succeed at challenging tasks, (2) making a positive attribution (optimism) about succeeding now and in the future, (3) persevering toward goals and when necessary redirecting paths to goals (hope) to succeed and (4) when beset by problems and adversity, sustaining and bouncing back and even beyond (resilience) to attain success. [P.3].

Therefore, and based on the proposed model, it is assumed that developing the psychological capital would impact the SMEs employees' innovative behavior in Saudi Arabia.

Companies desiring to innovate and offer novel products and services need to work on positive psychological capital among their employees. It is proved to positively impact different aspects related to employees, such as commitment and job satisfaction (Tang et al., 2019). There is also a necessity to consider practical measures for employees' psychological capital as this would ultimately lead to improving their overall performance and enhancing their innovative behavior. Having a positive psychological capital, particularly a high level of self-efficacy, increases the individual's capability to face challenges during the innovation processes (Tierney & Farmer, 2002). Hope, optimism, resilience, and job satisfaction were also reported to positively connect with work performance (Youssef & Luthans, 2007) and job efficiency.

Overall, it was reported that the better the psychological capital is, the higher job satisfaction is achieved. However, the constructs of psychological capital may be explained separately. Self-efficacy is defined as the confidence level obtained by an individual that, if increased, will lead to more job satisfaction. Individuals' belief in their capabilities to complete tasks has positive predictive power on interpersonal relationships, individuals' learning outcomes, and work performance (Judge et al., 2007; Zimmerman, 2000). Individuals with high self-efficacy levels believe in their abilities to mobilize the motivation, cognitive resources, and actions needed for successful performance (Stajkovic & Luthans, 1998). Because these individuals believe in their capabilities, they are most likely to choose challenging duties and tasks, use their motivational resources and efforts to perform their objectives, and defeat any obstacles that may hinder them (Luthans & Youssef, 2004). When individuals enjoy a high level of self-efficacy, they are believed to have

more satisfaction levels and behave innovatively within their enterprises.

The second construct of the psychological capital is hope, rooted originally in hope theory (Snyder, 2002). Hope is individuals' belief about their determination to achieve their goals and search for good ways to defeat the challenges and obstacles they may face during work or life activities. They also tend to face risks and select novel pathways when their current paths are blocked (Snyder, 2002). Once employees achieve a high degree of hope, they are likely to be happier and satisfied with their organizations (Peterson & Luthans, 2003). Therefore, this may make them intend to be a better inventor and be more creative than others. The optimism construct of psychological capital is also a critical factor in psychological theory. Previous investigations stated that optimism and hope have distinct and predictive powers (Bailey et al., 2007).

Optimism requires a precise evaluation of the present situations and an understanding of what can be done in such cases. Optimists create an exciting life, strengthening their self-esteem that may, in turn, lead to greater creativity (Goldsmith & Matherly, 1988). They also space themselves from unfavorable life events, minimizing the likelihood of facing depression, self-blame, guilt, and despair (Rego et al., 2012). As a result, their chances of giving up are meager. They enjoy more positivity, experience more positive emotions, have high perseverance when facing a challenge, always look for creative ways to solve problems, and positively benefit from opportunities (Youssef & Luthans, 2007).

Resilience is another essential factor in the umbrella of psychological capital. It is the ability to recover from any setback or adversity and adapt again to the business environment. Individual resilience is beneficial for them in their job-seeking skills (Fleig-Palmer et al., 2009). The resilient employees enjoy energetic approaches to life and are curious and open to new experiences (Tugade et al., 2004). Resilient employees can adapt to recent changes; develop new ways of dealing with challenges, difficulties, and failures; and recover from malicious emotional activities. Therefore, we in this study apply the principles of the psychological capital in the stream of organizational behavior to understand the psychological strengths of SME employees as they were believed to act as a mental source for developing human capital and social capital.

It is to be concluded from the above discussion that psychological capital may play a vital role in developing the employees' characteristics and capabilities that may positively affect their performance, satisfy them with their job, and motivate them toward innovation and creativity. For that reason, it seems interesting to investigate the relationship between psychological capital, job satisfaction, and innovative intention and whether job satisfaction and innovative intention act as mediators for the relationship or not. In addition, it is also motivating to investigate the direct impact of the psychological capital on the employees' innovative behavior. Thus, the following hypotheses are developed:

**Hypothesis 1:** Psychological capital has a positive influence on the employees' job satisfaction.

**Hypothesis 2:** Psychological capital has a positive influence on the employees' innovative intention.

**Hypothesis 3:** Psychological capital has a positive influence on the employees' innovative behavior.

## Employees' Job Satisfaction

Employees are considered a valuable asset and a ladder for building enterprises. Their satisfaction in general and job satisfaction, in particular, is a matter of concern for enterprises as they understand its significance and effect on their overall growth and development. The employees' satisfaction concept is vast: Robbins et al. (2015) define it as "the feelings or affective responses of a worker regarding factors such as working environment, working experience and the job itself" (as cited in Tang et al., 2019). It is that status in which employees feel satisfied with their current work and how it is performed and supervised. The level of employees' job satisfaction could be influenced by many factors such as the opportunities for decision-making participation, coworkers' support, supervisors' support, training chances, and workplace tension (Griffiths et al., 2011).

The hostile work environment may not just limit the employees from carrying out their routine activities. It would further discourage them from thinking innovatively and creatively toward developing novel products and services beneficial for their enterprises. Ensuring an acceptable level of work satisfaction among employees is very important although it is not the only factor determining the individuals' behavior in business organizations. The satisfaction gap that has resulted from the differences between what is expected by the employees and what is available for them needs to be filled to ensure full utilization of the available human resources.

When an organization hosts employees with a high level of payment, proper supervision, adequate promotion, cooperative coworkers, and work nature, this doubtlessly impacts their job satisfaction level and tightening their intention to stay longer (Alshetri, 2013). Moreover, it would further alter and positively direct their behavior, ultimately influencing their well-being and organizational performance (Spector, 1985). It may also pressure employees to comply with some standards when working that professionalizes them and allows them to think creatively.

Satisfied individuals can deliver excellent services and products, ultimately resulting in better outcomes and higher customer satisfaction and loyalty (Eskildsen & Dahlgard, 2000). Ensuring maximum satisfaction level and best utilization of available resources is always recommended, and thus there is a need for good work assigning among employees based on their specializations (Altaweel, 2019). Because employees are affected by various factors, thus there is a necessity to benefit from their psychological capital. Employees with a high level of hope, self-efficacy,



resilience, and optimism perform better than those with a low level.

Psychological capital, particularly self-efficacy, is believed to positively impact individuals in general and their satisfaction level in particular (Klassen & Chiu, 2010). Thus, it assists employees to balance between the satisfaction and dissatisfaction status and have overall satisfaction with their job. Furthermore, those employees with a high level of psychological capital tend to have a high level of confidence, ability to face challenges, determination to achieve their goals, and cheerfulness about the present and the future, which at last directs individuals to create a happy life and satisfying workplace.

Consequently, this establishes innovative behavior toward new products and services for the organizations' sustainability. Therefore, enterprises, regardless of their type and size, need to take innovative measures to ensure competitiveness and enhance their performance (Damanpour et al., 1989; Luoh et al., 2014). The above review proposes that the employees' job satisfaction may mediate between the psychological capital and the employees' innovative behavior. Also, there could be a direct positive relationship between employees' job satisfaction and innovative behavior. Thus, the following hypotheses are developed:

**Hypothesis 4:** The employees' job satisfaction plays a mediating role between the psychological capital and the employees' innovative behavior.

**Hypothesis 5:** The employees' job satisfaction has a positive relationship with the employees' innovative behavior.

## Employees' Innovative Intention

For an individual to carry out an activity or to perform a behavior, there has to be a hidden internal motivation to push them forward, which is the so-called intention. It has been discussed previously in various literature reviews. The top of them was the theory of planned behavior (TPB) developed by Ajzen (1991), who argued that attitude, subjective norms, and the perceived behavioral control influence human behavior. In this theory, intention was defined as the internal preparation people have toward a particular behavior. Accordingly, individuals should have the opportunity to enhance and use their knowledge as it will allow them to broaden their cognitive level.

Because innovation is the only source of competitive advantage, enterprises must have a specific intent. It will help their employees see how innovation connects to what they do and how they will be doing (Kaplan, 2017). The innovation intent of the enterprises and their members should concentrate on offering something different for targeted clients. Enterprises that operate without innovation intent tend to lose focus and miss their goals.

Employees need to be guided on what to do and what specific innovations are required from them. Thus, the lack of thoughtfully defined innovation intent makes it challenging for leaders to create specific programs, processes, matrices, and rewards that shape values and behavior (Kaplan, 2017). Thus, enterprises need to take adequate measures to stimulate innovation willingness among their employees and enhance their innovative behavior (Li & Zheng, 2014).

Employees should be provided with the required resources and opportunities so that their innovative intention is improved and innovative behavior is pushed. Enterprises also should work on the employees' psychological capital represented by its four constructs as it has a connection with the entrepreneurial intention, particularly the individual's self-efficacy and resilience (Contreras et al., 2017). Self-efficacy is confirmed to act as a predictor for entrepreneurial intention (Mat et al., 2017). Those employees with a high level of hope may also have a positive intention toward their job and invent new products and services beneficial for their enterprises. Therefore, employees' intention and their self-efficacy may have many factors affecting them as the psychological capital is about one's belief and the perception of one's ability (Caza et al., 2010). It is then expected that intention influences the performance to the extent that the person has behavioral control. Performance should increase with the behavioral control to the extent that the person is motivated to try (Ajzen, 1991).

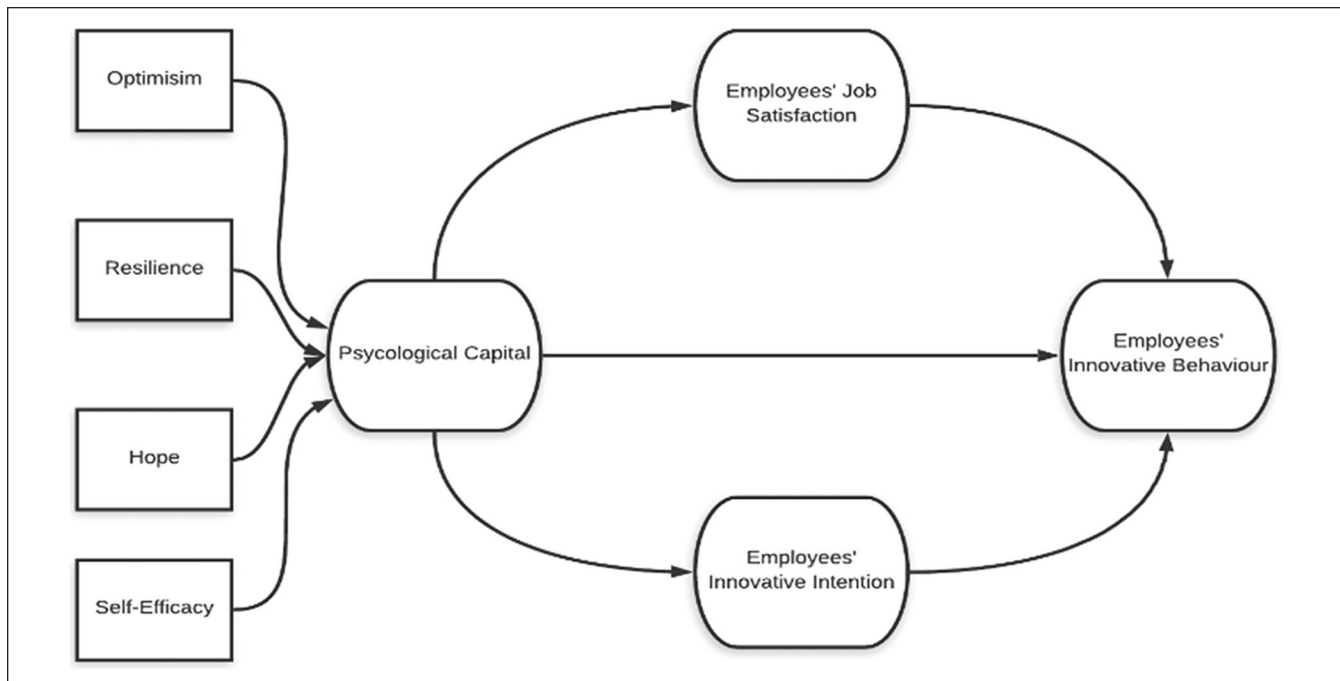
The intention is believed to be an accurate predictor of planned behavior; individuals with high entrepreneurial potential have more entrepreneurial intentions. Furthermore, optimism being part of the psychological capital is also believed to predict leadership behavior and intention (Chemers et al., 2000). To sum up, having a high level of psychological capital encourages individuals to focus more on the gain, persist in the face of adversities, and have an innovative intention toward innovative behavior. Thus, from the above discussion, it seems exciting to investigate whether the employees' innovative intention will play a mediating role between the psychological capital and their innovative behavior, and accordingly, the following hypotheses are developed:

**Hypothesis 6:** The employees' innovative intention mediates the relationship between psychological capital and innovative behavior.

**Hypothesis 7:** The employees' innovative intention has a positive relationship with the employees' innovative behavior.

## Employees' Innovative Behavior

For enterprises to sustain and continue operating effectively, there is a requirement for continuous improvement and development of their products and services that would



**Figure 1.** The hypothesized model.  
Source. Author's elaboration.

necessitate a joint effort from all their members. For that, enterprises need to take measures that enhance creativity and innovation processes among their employees. Innovation is one of the critical sources for generating a competitive advantage that helps in the market. Therefore, enterprises have always attempted to evoke creativity among their members to encourage creative ideas (Anderson et al., 2014). Innovative behavior is defined as developing, finding support, and implementing new ideas (Scott & Bruce, 1994). It is also stated as introducing new attractive ideas with responsibilities among employees group (Janssen, 2000) that are necessary for improving the efficiency and performance of the job (Baer & Frese, 2003). Those employees with higher self-confidence are more likely to conduct innovative organizational tasks (Tierney & Farmer, 2002). For that, high-tech companies have recently started focusing on recruiting talented employees who have the skills and abilities to behave innovatively and implement adequate measures to ensure a comfortable and attractive business environment for their individuals.

In addition to that, the employees' innovative behavior may be encouraged by the climate and culture of the organizations, job characteristics, social groups, individual differences, expected payoffs, and other factors. Although individuals' innovative behavior is an expression of individuals' ability and contributes to the benefits, success, and long-term survival, it is considered a risky endeavor. It brings benefits and costs for employees beyond their sense of intrinsic enjoyment (Janssen et al., 2004).

To conclude, innovative behavior is the overall performance of employees in the process of creative searching, establishing, implementing, and successful realizing new technologies, processes, techniques, or products to generate useful services or products (Tang et al., 2019), which makes it a fascinating subject for investigation. As a result, this study evaluates the role of psychological capital on the employees' innovative behavior in Saudi Arabia, considering the employees' job satisfaction and the employees' innovative intention as mediating factors for this analysis.

## Conceptual Framework

The proposed model represented in Figure 1 highlights the concept and purpose of the study. In this model, the psychological capital represented by optimism, resilience, hope, and self-efficacy is assumed to be the independent variable. The other side of the model represents the dependent variable, that is, employees' innovative behavior. Concerning the mediating effects, the employees' job satisfaction and innovative behavior were assumed to mediate between the independent and dependent variables.

## Method

### Participants and Data Collection

The primary data were collected with the help of an online questionnaire sent to the employees working in SMEs of

**Table 1.** Demographic Information of the Respondents.

| Characteristics              | Frequency | %   |
|------------------------------|-----------|-----|
| Gender                       |           |     |
| Male                         | 178       | 87  |
| Female                       | 26        | 13  |
| Total                        | 204       | 100 |
| Age                          |           |     |
| 18–29                        | 45        | 22  |
| 30–41                        | 78        | 38  |
| 42–53                        | 61        | 30  |
| Above 53                     | 20        | 10  |
| Total                        | 204       | 100 |
| Experience                   |           |     |
| Less than 10 years           | 108       | 53  |
| 10–20 years                  | 55        | 27  |
| Above 20 years               | 41        | 20  |
| Total                        | 204       | 100 |
| Sector                       |           |     |
| Real estate & construction   | 18        | 9   |
| Manufacturing                | 20        | 10  |
| Services                     | 67        | 33  |
| Production                   | 27        | 13  |
| Wholesale and retail         | 35        | 17  |
| Finance and insurance sector | 22        | 11  |
| Others                       | 15        | 7   |
| Total                        | 204       | 100 |

Source. Primary data.

various sectors in Saudi Arabia. In addition, other information was gathered from multiple sources such as books, previous studies, articles, reports, and other relevant materials. Because the study aimed to investigate the influence of the psychological capital on the employees' innovative behavior working in Saudi Arabia with mediating effect of job satisfaction and innovative intention, the study was not limited to a specific sector in the country. Therefore, the received valid responses for analysis totaled 204. The questionnaire was first sent to 13 employees from different enterprises to investigate any challenges with the questionnaire's measures, and nothing was reported. Then, the survey was sent and remained online for 1 month.

### Demographic Information

Table 1 describes the demographic information of the respondents.

### Measures

For measuring the constructs of the study, the following measures were employed as below:

Table 2 describes the sources of the measures of the study.

**Table 2.** Sources for Study's Measures.

| Measure                         | Source                              |
|---------------------------------|-------------------------------------|
| Psychological capital           | Luthans, Youssef, and Avolio (2007) |
| Employees' satisfaction         | Kiffin-Petersen and Corderly (2003) |
| Employees' innovative intention | Liñán and Chen (2009)               |
| Employees' innovative behavior  | Scott and Bruce (1994)              |

Source. Author's elaboration.

### Data Analysis

#### Measurement Model

The study used the SmartPLS for analysis and testing of the hypotheses. It uses a nonparametric tactic in assessing the psychometrics of the scales and the path coefficients. And it has a smaller amount of restrictions on the linearity, normality, and sample size (Hair et al., 2012).

The reliability and validity were tested using four different methods: the factor loading, Cronbach's alpha, composite reliability, and average variance extracted (AVE). According to Chew et al. (2019), the values of the variables in factor loading found to be less than 0.7 should be avoided unless they are essential for validity contents. Thus, the threshold value was determined as 0.6.

Table 2 results indicate that most of the items are higher than 0.7, except for two items. Cronbach's alpha, in which the study of Vaske et al. (2017) has indicated that it measures the constructs' internal consistency, counts the reliability. The constructs' reliability is achieved when Cronbach's alpha and composite reliability are more significant than .70. If variables are detected to be lower than .6, the internal consistency among variables is weak. As per Cronbach's alpha results in Table 3, the study variables have a value above .6, indicating that they are dependable and can be used for examination.

Another measure for reliability is composite reliability, in which the threshold is considered to be .6. Composite reliability results of all the variables show a value above the .6 thresholds indicating the composite reliability. The last element examined is the AVE and has a threshold of 0.5 as per Shau (2017). Therefore, all the constructs have a value of AVE above 0.5, which meets the point. Hence, it is proven from the results that the variables are reliable and valid, which can be used for conducting further tests.

Table 3 reveals constructs reliability and validity for the variables used in the study.

Table 4 reveals the discriminate validity among the various constructs of the study.

Table 5 discloses the Fornell–Larcker Criterion among the variables of the study.

**Table 3.** Constructs Reliability and Validity.

| Constructs                      | Factor loadings | Cronbach's $\alpha$ | Composite reliability | Average variance extracted |
|---------------------------------|-----------------|---------------------|-----------------------|----------------------------|
| Employees' innovative behavior  |                 | .763                | .849                  | 0.586                      |
| Item 2                          | 0.768           |                     |                       |                            |
| Item 3                          | 0.687           |                     |                       |                            |
| Item 4                          | 0.848           |                     |                       |                            |
| Item 5                          | 0.751           |                     |                       |                            |
| Hope                            |                 | .728                | .830                  | 0.550                      |
| Item 2                          | 0.728           |                     |                       |                            |
| Item 4                          | 0.734           |                     |                       |                            |
| Item 5                          | 0.739           |                     |                       |                            |
| Item 6                          | 0.765           |                     |                       |                            |
| Employees' innovative intention |                 | .848                | .891                  | 0.620                      |
| Item 1                          | 0.820           |                     |                       |                            |
| Item 2                          | 0.747           |                     |                       |                            |
| Item 3                          | 0.774           |                     |                       |                            |
| Item 4                          | 0.757           |                     |                       |                            |
| Item 5                          | 0.836           |                     |                       |                            |
| Optimism                        |                 | .769                | .853                  | 0.593                      |
| Item 1                          | 0.703           |                     |                       |                            |
| Item 3                          | 0.838           |                     |                       |                            |
| Item 4                          | 0.835           |                     |                       |                            |
| Item 5                          | 0.694           |                     |                       |                            |
| Resilience                      |                 | .640                | .844                  | 0.731                      |
| Item 2                          | 0.902           |                     |                       |                            |
| Item 5                          | 0.805           |                     |                       |                            |
| Employees' job satisfaction     |                 | .764                | .851                  | 0.590                      |
| Item 3                          | 0.655           |                     |                       |                            |
| Item 4                          | 0.722           |                     |                       |                            |
| Item 5                          | 0.867           |                     |                       |                            |
| Item 6                          | 0.811           |                     |                       |                            |
| Self-efficacy                   |                 | .756                | .845                  | 0.577                      |
| Item 2                          | 0.767           |                     |                       |                            |
| Item 3                          | 0.790           |                     |                       |                            |
| Item 4                          | 0.729           |                     |                       |                            |
| Item 5                          | 0.751           |                     |                       |                            |

Source. Primary data.

## Structural Model

### Hypotheses Testing

To test the study's hypotheses and to validate the study constructs, the bootstrapping procedure was performed for this purpose with 500 resamples.

As per the results shown in Table 6, the study's hypotheses relationships were not all supported. First, the innovative intention construct did not positively connect with the employees' innovative behavior construct ( $\beta = -0.008$ ,  $p = .890$ ). This confirms no positive connection between the employees' innovative intention and employees' innovative behavior. A logical reason might be that employees might have an intention but have no clue how to convert it into actual work due to not receiving the necessary supports.

Furthermore, the table shows a significant positive connection between psychological capital and the employees' innovative behavior ( $0.243$ ,  $p = .000$ ), which is expected as employees with more hope, self-efficacy, resilience, and optimism tend to act more effectively innovatively. Hence the hypothesis is accepted.

It was further revealed that the psychological capital has a significant positive relationship with employees' innovative intention ( $\beta = 0.482$ ,  $p = .000$ ), which is also anticipated as the intention is a result of factors that either promote or discourage individuals from conducting given behavior. Thus, the hypothesis is also supported.

Also, the connection between the psychological capital and the employees' job satisfaction was tested and reported as ( $0.421$ ,  $p = .000$ ), indicating a significant relationship. It is also an expected finding as those with a high level of hope,



**Table 4.** Discriminate Validity—Cross Loading.

|         | Employees'<br>innovative behavior | Hope  | Employees'<br>innovative intention | Optimism | Resilience | Satisfaction | Self-efficacy |
|---------|-----------------------------------|-------|------------------------------------|----------|------------|--------------|---------------|
| EIB2    | 0.768                             | 0.225 | 0.127                              | 0.274    | 0.098      | 0.562        | 0.148         |
| EIB3    | 0.687                             | 0.250 | 0.271                              | 0.398    | 0.321      | 0.368        | 0.226         |
| EIB4    | 0.848                             | 0.336 | 0.218                              | 0.380    | 0.142      | 0.517        | 0.284         |
| EIB5    | 0.751                             | 0.247 | 0.226                              | 0.342    | 0.149      | 0.477        | 0.276         |
| Hope2   | 0.218                             | 0.728 | 0.208                              | 0.405    | 0.213      | 0.203        | 0.287         |
| Hope4   | 0.269                             | 0.734 | 0.289                              | 0.400    | 0.283      | 0.347        | 0.340         |
| Hope5   | 0.273                             | 0.739 | 0.296                              | 0.449    | 0.485      | 0.182        | 0.364         |
| Hope6   | 0.263                             | 0.765 | 0.358                              | 0.428    | 0.275      | 0.303        | 0.264         |
| EII1    | 0.293                             | 0.384 | 0.820                              | 0.425    | 0.266      | 0.245        | 0.332         |
| EII2    | 0.152                             | 0.241 | 0.747                              | 0.387    | 0.317      | 0.189        | 0.250         |
| EII3    | 0.157                             | 0.292 | 0.774                              | 0.228    | 0.131      | 0.202        | 0.210         |
| EII4    | 0.210                             | 0.269 | 0.757                              | 0.312    | 0.233      | 0.295        | 0.204         |
| EII5    | 0.213                             | 0.324 | 0.836                              | 0.284    | 0.194      | 0.241        | 0.181         |
| Optim1  | 0.209                             | 0.332 | 0.225                              | 0.703    | 0.356      | 0.235        | 0.174         |
| Optim3  | 0.344                             | 0.517 | 0.392                              | 0.838    | 0.498      | 0.307        | 0.292         |
| Optim4  | 0.359                             | 0.418 | 0.382                              | 0.835    | 0.387      | 0.378        | 0.362         |
| Optim5  | 0.468                             | 0.469 | 0.294                              | 0.694    | 0.199      | 0.590        | 0.261         |
| Resi2   | 0.263                             | 0.484 | 0.301                              | 0.454    | 0.902      | 0.117        | 0.267         |
| Resi5   | 0.091                             | 0.215 | 0.194                              | 0.346    | 0.805      | 0.038        | 0.278         |
| Satisf3 | 0.346                             | 0.304 | 0.219                              | 0.413    | 0.282      | 0.655        | 0.202         |
| Satisf4 | 0.418                             | 0.217 | 0.307                              | 0.354    | 0.048      | 0.722        | 0.090         |
| Satisf5 | 0.555                             | 0.327 | 0.234                              | 0.423    | 0.011      | 0.867        | 0.081         |
| Satisf6 | 0.597                             | 0.224 | 0.178                              | 0.321    | −0.003     | 0.811        | 0.139         |
| Self2   | 0.253                             | 0.285 | 0.197                              | 0.323    | 0.202      | 0.168        | 0.767         |
| Self3   | 0.258                             | 0.269 | 0.232                              | 0.166    | 0.221      | 0.093        | 0.790         |
| Self4   | 0.185                             | 0.378 | 0.293                              | 0.244    | 0.254      | 0.131        | 0.729         |
| Self5   | 0.229                             | 0.349 | 0.215                              | 0.341    | 0.277      | 0.099        | 0.751         |

Source. Primary data.

**Table 5.** Fornell–Larcker Criterion.

| Construct                          | Employees'<br>innovative behavior | Hope  | Employees'<br>innovative intention | Optimism | Resilience | Employees' job<br>satisfaction | Self-<br>efficacy |
|------------------------------------|-----------------------------------|-------|------------------------------------|----------|------------|--------------------------------|-------------------|
| Employees' innovative<br>behavior  | 0.766                             |       |                                    |          |            |                                |                   |
| Hope                               | 0.346                             | 0.742 |                                    |          |            |                                |                   |
| Employees' innovative<br>intention | 0.269                             | 0.390 | 0.787                              |          |            |                                |                   |
| Optimism                           | 0.450                             | 0.568 | 0.428                              | 0.770    |            |                                |                   |
| Resilience                         | 0.221                             | 0.431 | 0.297                              | 0.475    | 0.855      |                                |                   |
| Employees' job satisfaction        | 0.635                             | 0.348 | 0.299                              | 0.488    | 0.098      | 0.768                          |                   |
| Self-efficacy                      | 0.304                             | 0.425 | 0.309                              | 0.360    | 0.316      | 0.163                          | 0.760             |

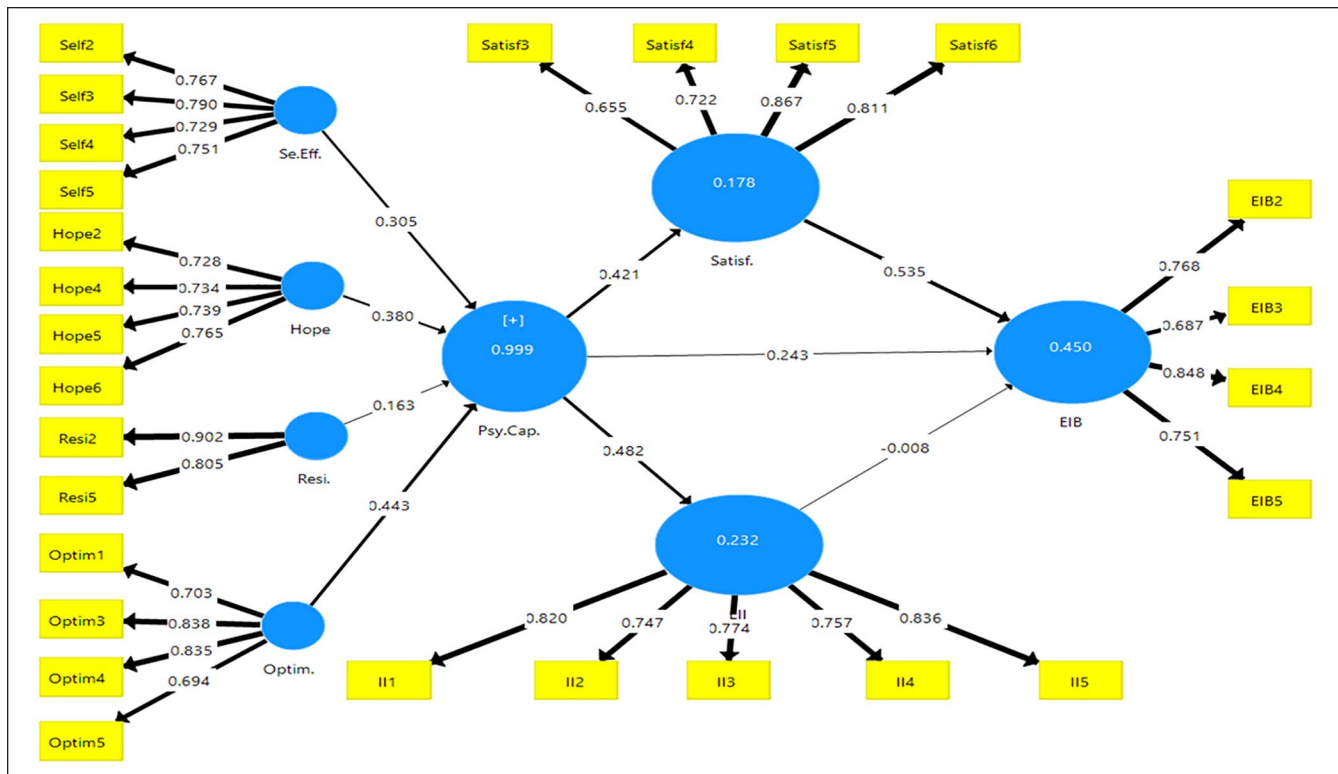
Source. Primary data.

optimism, self-efficacy, and resilience will like their jobs more than others who are always negative and have less trust in their enterprises. Accordingly, the alternative hypothesis was accepted.

Finally, the relationship between employees' job satisfaction and the employees' innovative behavior was also significant ( $\beta = 0.535$ ,  $p = .000$ ), and accordingly, the alternative

hypothesis is accepted. Therefore, those satisfied employees will tend to have better productivity and performance than dissatisfied ones.

The overall result of the study shows that the psychological capital, employees' satisfaction, and employees' innovative intention constructs predicted 45% of the variance in the employees' innovative behavior. It also reported that the



**Figure 2.** The result of the coefficients.

Source. Primary data.

**Table 6.** Path Coefficients.

| Constructs Relationships                                         | $\beta$ | <i>M</i> | <i>SD</i> | <i>T</i> statistics | <i>p</i> value | Decision |
|------------------------------------------------------------------|---------|----------|-----------|---------------------|----------------|----------|
| Employees' innovative intention → Employees' innovative behavior | -0.008  | -0.008   | 0.055     | 0.139               | .890           | Rejected |
| Psychological capital → Employees' innovative behavior           | 0.243   | 0.244    | 0.068     | 3.587               | .000           | Accepted |
| Psychological capital → Employees' innovative intention          | 0.482   | 0.489    | 0.059     | 8.120               | .000           | Accepted |
| Psychological capital → Employees' job satisfaction              | 0.421   | 0.429    | 0.070     | 6.049               | .000           | Accepted |
| Employees' job satisfaction → Employees' innovative behavior     | 0.535   | 0.538    | 0.064     | 8.420               | .000           | Accepted |

Source. Primary data.

psychological capital predicted 17.8% of the variance in the employees' satisfaction and 23.3% of the variance in the employee's innovative intention. Therefore, as predicting is higher than 10%, the model is accepted for prediction.

According to the results shown in Table 7 concerning the specific indirect effect for the study's constructs, the indirect impact of the relationship between psychological capitals → employees' innovative intention → employees' innovative behavior reported values of  $\beta = -0.004$ ,  $p = .892$ . Therefore, the alternative hypothesis is rejected and confirms that the employees' innovative intention does not mediate the relationship between psychological capital and its innovative behavior. The table further disclosed the relationship between psychological capital → employees' job satisfaction → employees' innovative behavior ( $\beta = 0.225$ ,  $p = .000$ ), indicating that the employees' job satisfaction partially mediates

the relationship between the psychological capital and the employees' innovative behavior.

### Construct Cross-Validated Redundancy

According to the ( $= 1 - [SSE/SSO]$ ) result shown in Table 8, all values of the constructs are greater than 0, confirming the model's adequacy to predict.

### Discussion

The study investigated the influence of psychological capital on the employees' innovative behavior working in the SMEs of Saudi Arabia. It employed the employees' job satisfaction and innovative intention as mediators between the original relationships. The study revealed that the employees' job

**Table 7.** Specific Indirect Effects.

| Specific Indirect Effects                                                                | B      | M      | SD    | T statistics | p value | Decision |
|------------------------------------------------------------------------------------------|--------|--------|-------|--------------|---------|----------|
| Psychological capital → Employees' innovative intention → Employees' innovative behavior | −0.004 | −0.004 | 0.027 | 0.136        | .892    | Rejected |
| Psychological capital → Employees' satisfaction → Employees' innovative behavior         | 0.225  | 0.230  | 0.045 | 4.998        | .000    | Accepted |

Source. Primary data.

**Table 8.** Construct Cross-Validated Redundancy.

| Constructs                      | SSO       | SSE     | $Q^2 (=1 - [SSE/SSO])$ |
|---------------------------------|-----------|---------|------------------------|
| Employees' innovative behavior  | 816.000   | 610.581 | 0.252                  |
| Employees' innovative intention | 1,020.000 | 887.861 | 0.130                  |
| Employees' job satisfaction     | 816.000   | 733.621 | 0.101                  |

Source. Primary data.

satisfaction plays a mediating role between psychological capital and innovative behavior. It further reported that the employees' innovative intention does not play a mediating role between the psychological capital and the employees' innovative behavior, emphasizing the need to develop new programs and courses necessary to enhance the employees' innovative intention in various sectors. The results of the direct relations were all mentioned in Table 6.

The study's findings were in line with Eskildsen and Dahlgard (2000) and Tang et al. (2019), showing the influence of psychological capital on employees' job satisfaction unlike the study of Idris and Manganaro (2017) that reported no significant relationship between psychological capital and employees' job satisfaction. Other studies such as Abosaif (2018), Aljaghtami and Noormala (2016), Ibrahim and Amari (2018), and Saad Alkahtani et al. (2020) also confirmed the positive impact of psychological capital on various constructs. Extant literature has also revealed psychological capital is narrowly related to their work performance (Avey et al., 2008), positivity, and creativity (Rego et al., 2012).

## Implications

The research contributes to the available limited literature about psychological capital globally, particularly in the Saudi Arabia context. It sheds light on one of the critical topics that contribute to improving the employees' performance in general and innovative. It provides an empirical evidence on the effect of the psychological capital in enhancing the higher level of satisfaction, developing innovative intention, and accordingly developing innovative products and services for their enterprises through strengthening the employees' abilities and internal traits that might help Saudi government increases the contribution of the SMEs from 20% to 35%.

Moreover, SME decision-makers need to focus on improving the employees' innovative performance by developing

their psychological capital (hope, resilience, optimism, and self-efficacy) and directing them toward their jobs. They, for example, need to ensure that employees have a high level of confidence in themselves and the ability to deal with uncertainties and challenges. They also have to make their employees think positively about success and achieve higher positions, motivating them to work innovatively and accordingly meet their objectives. They also need to ensure the provision of necessary innovative environment and infrastructures, which provides a higher level of the employee satisfaction and assists in directing their innovative attitude and behavior. Those employees with innovative behavior need to be encouraged and supported, and rewarded financially, technically, and emotionally to ensure their enterprises' continuous achievement and development. Others who are otherwise should be provided with the necessary training and programs to ensure improving their performance as per the desired one and consequently can contribute to the overall plan of the Saudi vision 2030. On the contrary, the government and policymakers should continue the provision of the necessary financial and technical support and training for the SMEs sector to allow further development of the available employees and permit recruiting of those skilled and innovative individuals. It should also direct their effort toward eliminating those challenges that may hinder the development of new enterprises or hamper the growth of the existing ones. In addition, there is a need to reduce the rate of tax, loan interest rate, and lengthy enterprises' establishment procedures to ensure rapid growth in the sector.

## Limitations and Future Research

The research targeted a sample of SME employees operating in Saudi Arabia, meaning there could be differences in understanding and responses for the study measures. Furthermore,

the received sample was limited, which may hamper its generalizability. The study also did not investigate the effect of control variable such as experience, age, and other. Future research may target more samples collected from specific sectors in the country and analyze the impact of psychological capital on the employees' burnout or other related aspects. There might also be a chance to compare the concept of psychological capital in Saudi Arabia with different contexts worldwide.

## Conclusion

Competition and globalization have become two of the critical challenges facing business enterprises these days. It necessitates them to think of alternative ways to ensure sustainability and generate a competitive advantage for them. Developing a competitive advantage requires a joint effort from various organizations and essential resources such as adequate capital, good quality material, and skilled human resources. Competitive advantage is obtained only if the company can develop new products and services distinctive from others, helping the business organizations bring a significant market share. This will be achieved if recruited employees are capable of thinking productively and acting innovatively. Employees' innovation and creativity are essential features for companies' sustainability and development. Innovative employees help companies in cost reduction, efficient operations, and effort minimization. Therefore, the employees' innovative behavior needs to be encouraged and supported by various means, including providing a suitable workplace and offering the required financial and nonfinancial incentives.

There is also a need to pull their inner innovative intention toward innovation and creativity, which may require more on-the-job training and development. Accordingly, the study considers psychological capital as an essential factor in pushing the employees' performance forward and improving their overall performance and attitude in the workplace. Thus, the study evaluated whether the psychological capital represented by hope, self-efficacy, optimism, and resilience influence the overall innovate behavior of the employees through the mediation effect of satisfaction and innovative intention and disclosed interesting results. The study recommends the focus on enhancing employees' positive psychological capital to ensure maximum benefits for the enterprises through the development of innovative products and services. It also increases their confidence level in achieving goals and facing challenging adversities. Employees with high-level psychological capital will have more creativity, hope, and optimism.

## Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

## Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: The author acknowledges the Deanship of Scientific Research at King Faisal University of the Financial Support under Nasher Track (Grant No. 206122).

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